Multimodal Embedding Audit

Embeddings file: /net/mraid20/ifs/wisdom/segal_lab/genie/LabData/Analyses/adamgab/code
N visits: 240
N subjects: 120
N baseline->02_00 pairs: 120

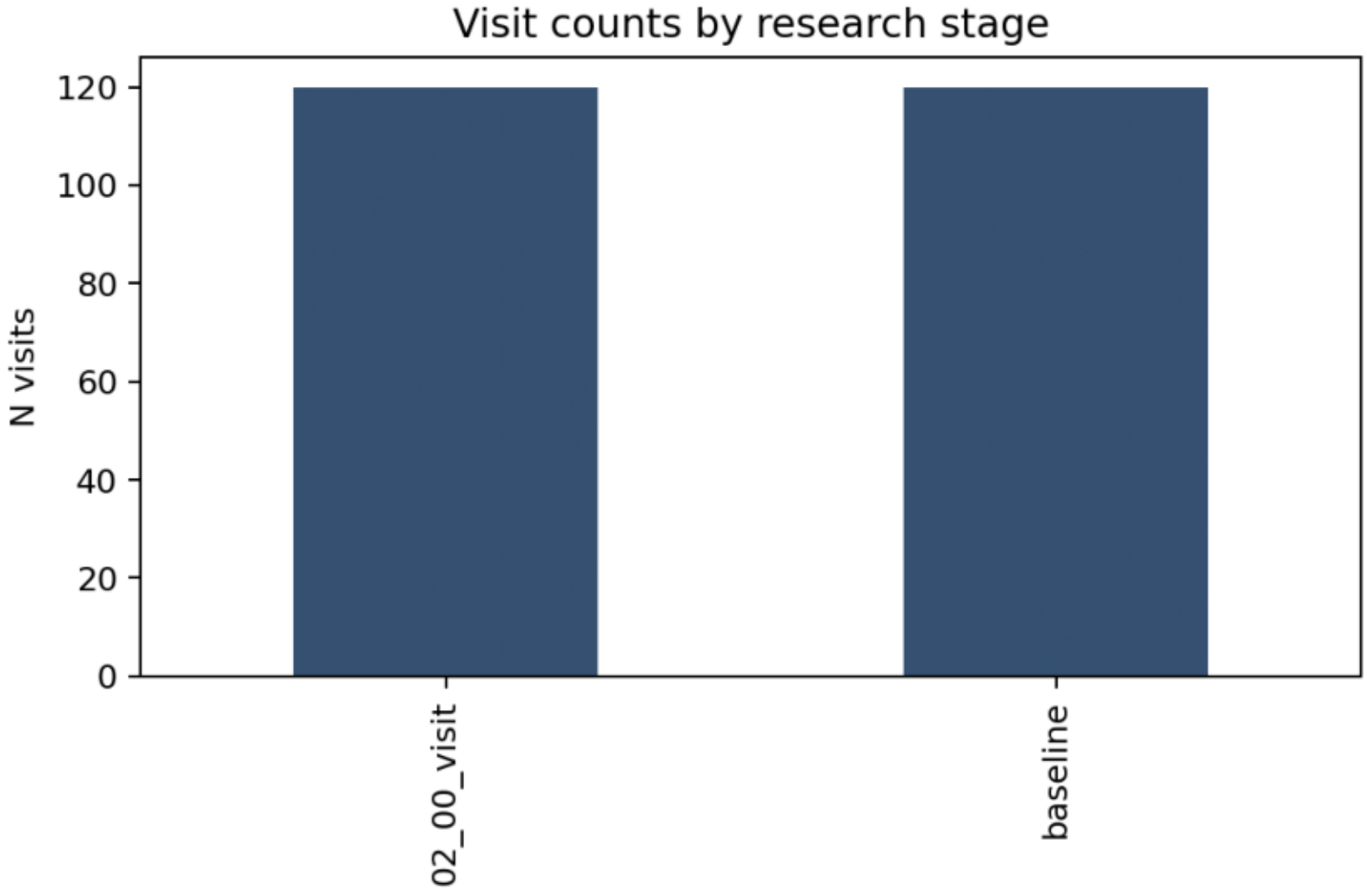
EVAL_SCORE (vs baseline): -0.0774
Task 1 mean r: 0.2105
Task 2 mean r: 0.2502

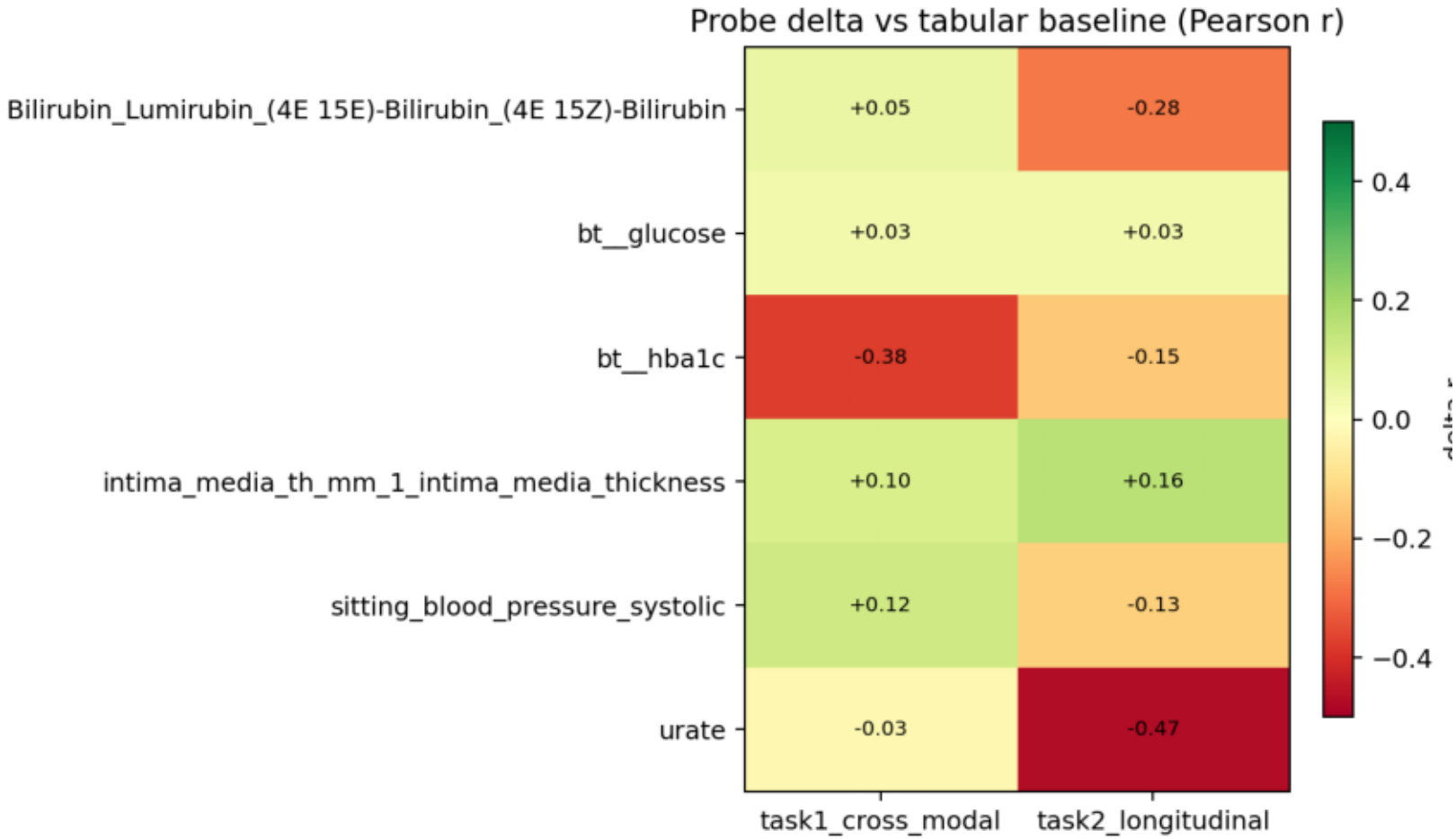
After rerun benchmark + AGB predictors: EVAL_SCORE=-0.0764
Task1=0.2125, Task2=0.2502

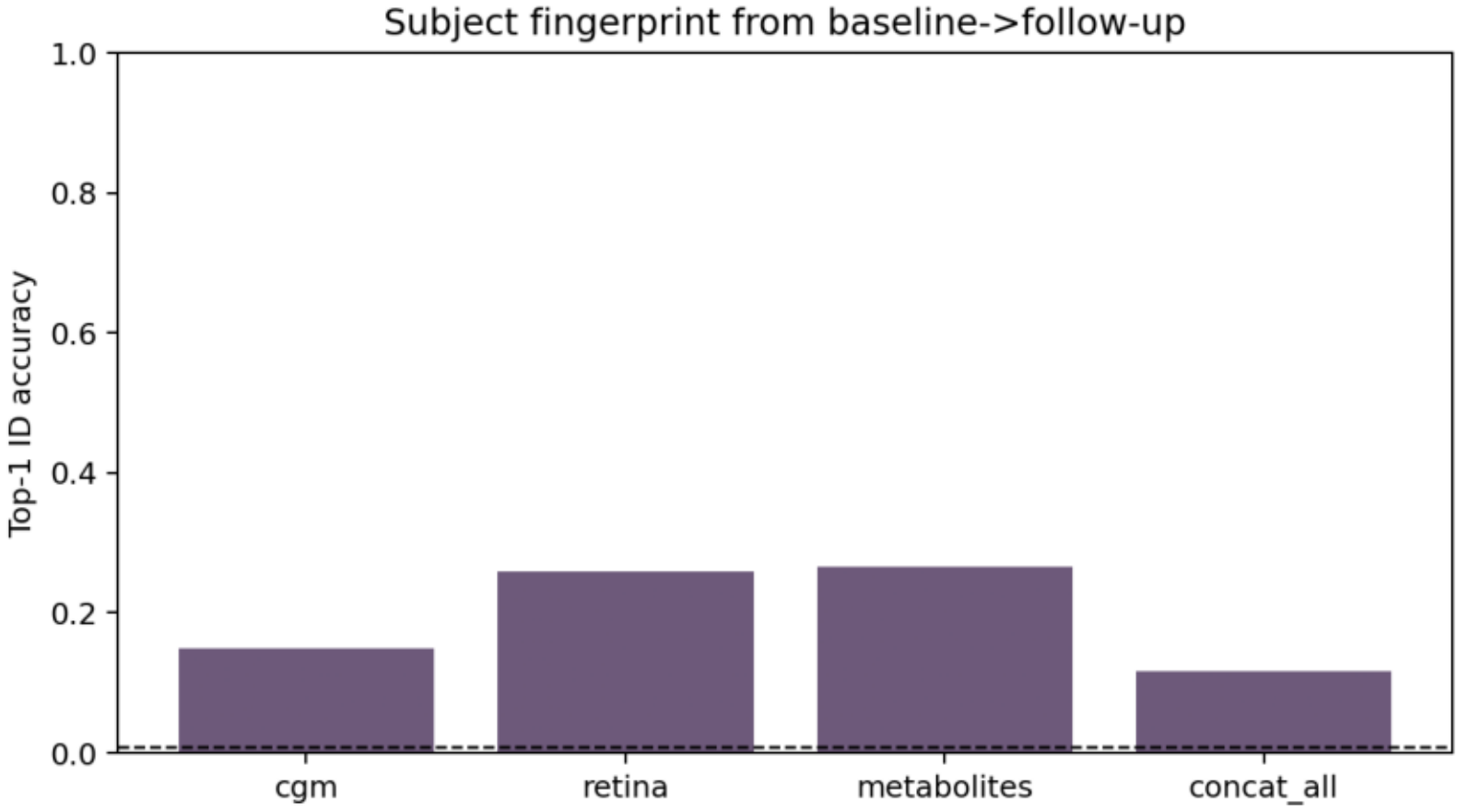
Top KPI deltas (probe - baseline):
- task2_longitudinal | intima_media_th_mm_1_intima_media_thickness: +0.161
- task1_cross_modal | sitting_blood_pressure_systolic: +0.118
- task1_cross_modal | intima_media_th_mm_1_intima_media_thickness: +0.100
- task1_cross_modal | Bilirubin_Lumirubin_(4E 15E)-Bilirubin_(4E 15Z)-Bilirubin: +0.054
- task2_longitudinal | bt__glucose: +0.034
- task1_cross_modal | bt__glucose: +0.034

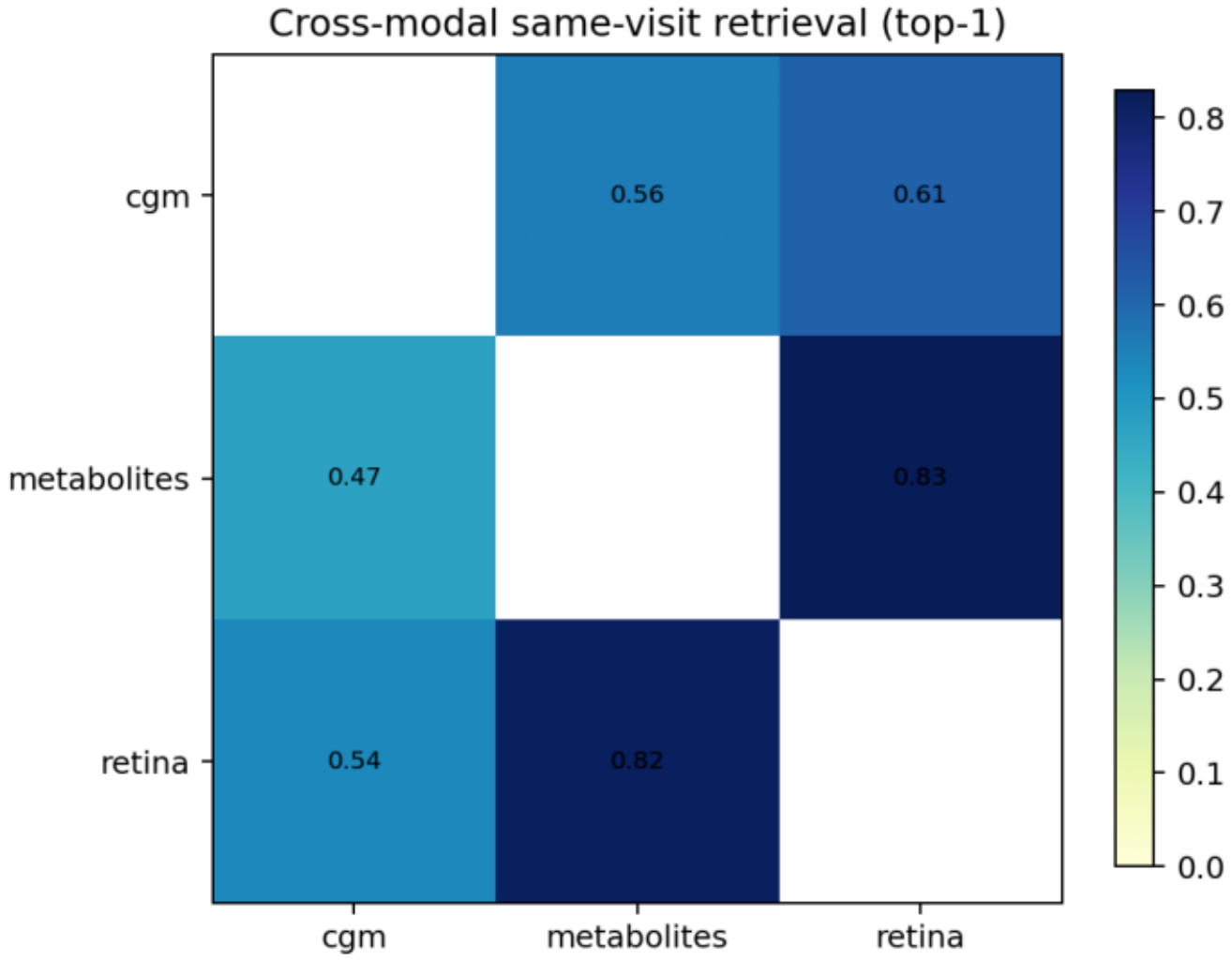
Selected novel probes (embedding+AGB - embedding):

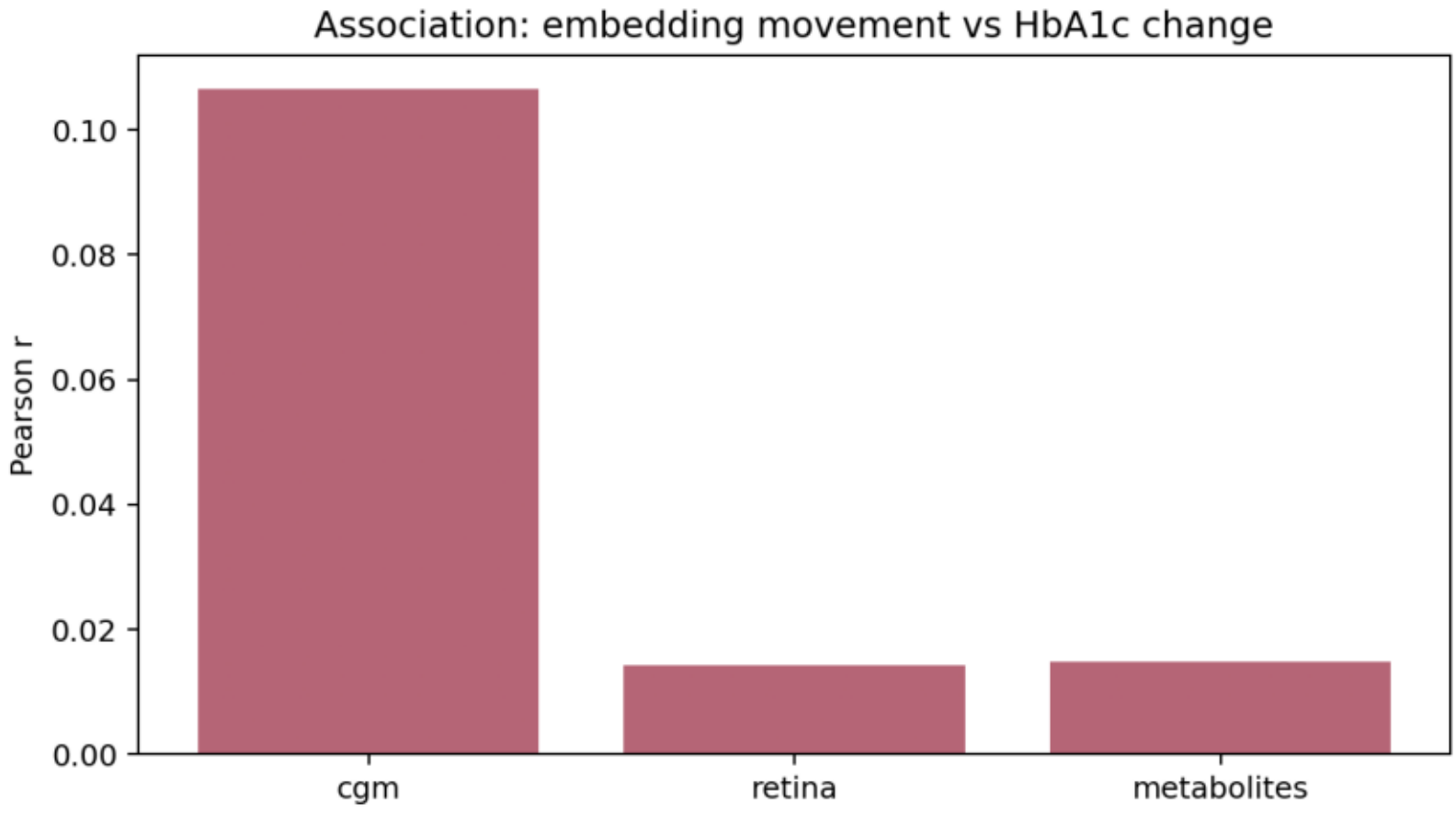
- body_comp_total_lean_mass (dexa lean mass): r=0.851 (emb=0.850, gain=+0.000, n=240)
- total_scan_vat_area (dexa visceral fat): r=0.800 (emb=0.799, gain=+0.001, n=240)
- iglu_mage (cgm excursions): r=0.741 (emb=0.740, gain=+0.001, n=102)
- iglu_cv (cgm variability): r=0.723 (emb=0.723, gain=+0.000, n=102)
- iglu_hbgi (cgm hyper risk): r=0.708 (emb=0.707, gain=+0.001, n=102)
- qrs_ms (cardio electrical): r=0.332 (emb=0.332, gain=+0.000, n=239)
- hr_bpm (cardio resting rate): r=0.265 (emb=0.264, gain=+0.001, n=240)
- total_scan_sat_mass (dexa adiposity): r=-0.146 (emb=-0.151, gain=+0.005, n=240)

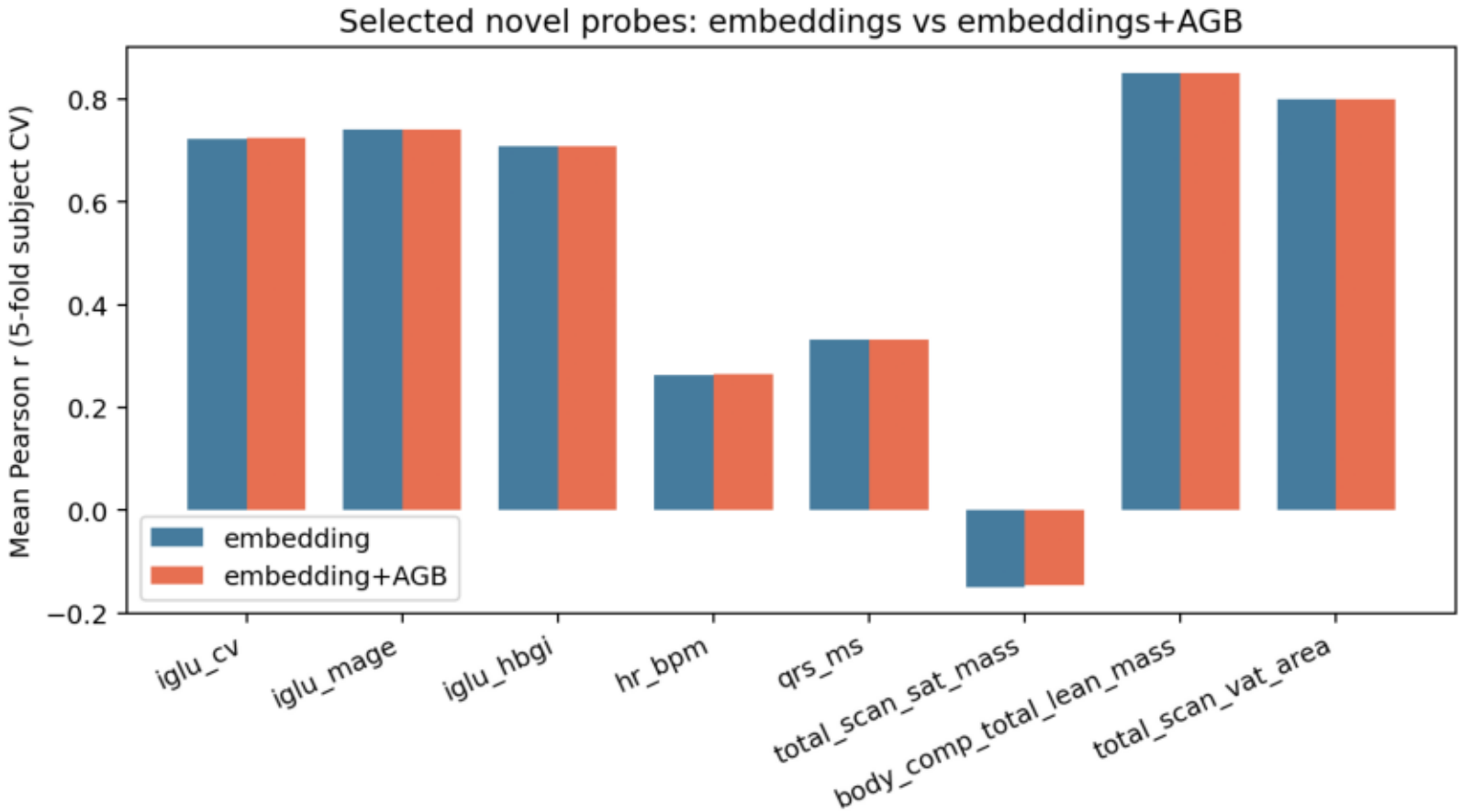

Visit counts by research stage
N visits
0
20
40
60
80
100
120
02_00_visit
baseline


Probe delta vs tabular baseline (Pearson r)
Bilirubin_Lumirubin_(4E 15E)-Bilirubin_(4E 15Z)-Bilirubin
bt__glucose
bt__hba1c
intima_media_th_mm_1_intima_media_thickness
sitting_blood_pressure_systolic
urate
+0.05
-0.28
+0.03
+0.03
-0.38
-0.15
+0.10
+0.16
+0.12
-0.13
-0.03
-0.47
task1_cross_modal
task2_longitudinal
0.4
0.2
0.0
−0.2
−0.4


Subject fingerprint from baseline->follow-up
Top-1 ID accuracy
1.0
0.8
0.6
0.4
0.2
0.0
cgm
retina
metabolites
concat_all


Cross-modal same-visit retrieval (top-1)
cgm
metabolites
retina
0.56
0.61
0.47
0.83
0.54
0.82
cgm
metabolites
retina
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0


Association: embedding movement vs HbA1c change
Pearson r
0.10
0.08
0.06
0.04
0.02
0.00
cgm
retina
metabolites


Selected novel probes: embeddings vs embeddings+AGB
Mean Pearson r (5-fold subject CV)
0.8
0.6
0.4
0.2
0.0
−0.2
embedding
embedding+AGB
iglu_cv
iglu_mage
iglu_hbgi
hr_bpm
qrs_ms
total_scan_sat_mass
body_comp_total_lean_mass
total_scan_vat_area